

Brain MRI Tumor Segmentation

Milestone 1 Report

Data Acquisition, Quality Control, Dataset Harmonization,
Splitting, Standardization, Model-Input Smoke Testing, Metrics, and Experiment
Tracking

Project Name	Brain MRI Tumor Segmentation
Milestone	Milestone 1 - Data Preparation, Standardization, and Model-Input Smoke Validation
Program	DEPI — Machine Learning Engineering
Technical Basis	Shared Colab/Jupyter notebook and team handoff instructions
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This report records the work represented in the supplied project notebook and the accompanying team handoff instructions.

It describes the implemented workflow without introducing unverified project results.

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1 Executive Summary

Milestone 1 establishes the data foundation for the **Brain MRI Tumor Segmentation** project. The work represented in the supplied notebook covers acquisition and organization of several brain-MRI datasets, metadata consolidation, file-level quality control, checksum verification, geometric inspection, label inspection, duplicate-image auditing, leakage-safe dataset splitting, image standardization, processed-data auditing, and packaging of the final standardization logic into a reusable pipeline.

The final workflow transforms heterogeneous MRI volumes into a common representation suitable for downstream model development. The saved pipeline converts images to a canonical RAS orientation, targets a common volume size of $240 \times 240 \times 155$, applies brain-region z-score normalization to imaging modalities, and applies the documented segmentation-label remapping where configured. The pipeline is intentionally restricted to development data for routine regeneration: its default `regenerate()` configuration uses BraTS2021, MU, and RHUH with only train and val splits, while test and external remain locked for final evaluation.

The notebook records the completion of a large standardization run with **10,563 processed NIfTI files**, of which **10,363 were newly written and 200 were already present and therefore skipped**. The resulting processed data occupied approximately **36.26 GB**. A subsequent audit found all 10,563 processed files at the target shape and RAS orientation. A 40-file comparison against a previously generated processed state reported **0 mismatches**.

Milestone 1 also includes the first end-to-end model-input smoke test (M1-T05). The implemented loader, MONAI 3D U-Net interface, augmentation pipeline, validation path, metric functions, and MLflow experiment tracking were executed on a deliberately small BraTS2021 subset. The smoke run used 16 training patients, 4 validation patients, 2 epochs, a 128^3 training patch, batch size 2, and CPU execution. The run completed successfully and produced both a JSON report and a model checkpoint. These values are explicitly treated as implementation-validation evidence rather than final model performance.

Milestone 1 outcome: the project reached a reproducible, QC-backed preprocessing state in which the standardized data and the saved pipeline can be handed to the downstream modeling stage without rebuilding the preprocessing logic from scratch.

2 Milestone 1 Objectives

The milestone was structured around six technical objectives:

1. Acquire or organize the required brain-MRI datasets and their supporting metadata.
2. Build a unified index describing subjects, timepoints, modalities, labels, provenance, QC state, and data paths.
3. Verify file existence, integrity, readability, shape, spacing, and orientation before standardization.

4. Prevent data leakage by auditing duplicate image files and generating a locked train/validation/test/external split.
5. Standardize the imaging volumes into a common coordinate system, shape, intensity space, and label representation.
6. Save the preprocessing logic as a reusable pipeline and validate its reproducibility and output consistency.
7. Connect the standardized data to the downstream model input pipeline and verify that an end-to-end smoke training run executes correctly.
8. Introduce experiment tracking through MLflow and validate the implemented segmentation metrics used by the downstream training/evaluation stages.

3 Overall Data Flow

The milestone can be summarized as the following sequence:

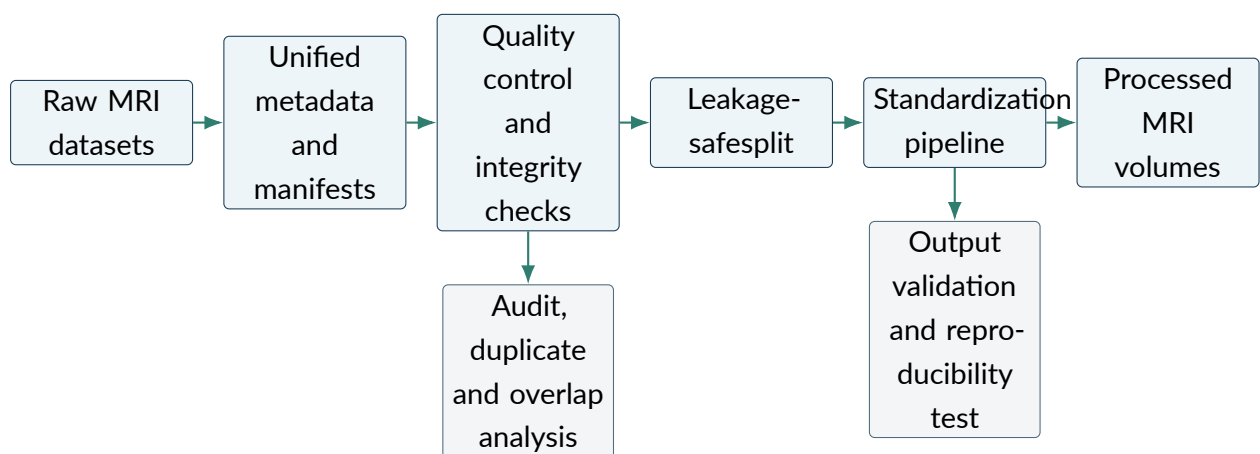


Figure 1: Milestone 1 end-to-end data-preparation workflow.

4 Data Acquisition and Initial Organization

The notebook begins by preparing a Google Colab environment and installing the tools required for dataset acquisition. The initial acquisition stage uses Kaggle for selected datasets and direct downloads for other public research datasets. The notebook contains acquisition steps for:

- BraTS 2021 Task 1.
- A BraTS 2024 glioma dataset source used during the acquisition stage.
- RHUH-GBM.
- BraTS-Africa.
- MU-Glioma-Post.

- UTSW-Glioma.
- OpenNeuro dataset ds007045.

The acquisition cells store files under `/content/data` during the Colab session and use the project Drive area `BrainMRI_Data` as the persistent project repository. ZIP/TAR extraction and deletion of temporary archives are used where appropriate to manage disk pressure.

Important implementation detail: the final reusable regeneration pipeline does not simply standardize every dataset that appeared in the initial acquisition notebook. Its default configuration is explicitly limited to `brats2021`, `mu`, and `rhuh` and to the `train` and `val` splits. This distinction separates exploratory/raw acquisition from the final reproducible development pipeline.

The persistent project structure established on Google Drive is organized around the following directories:

Listing 1: Project storage layout

```
1 BrainMRI_Data/|—
2   raw/|—
3   manifests/|—
4   qc/|—
5   pipeline/|—
6     standardize.py
```

The notebook also verifies disk capacity throughout the acquisition and extraction stages. The team handoff notes that the standardization process may require approximately 60 GB of free Colab runtime storage because source archives and generated processed data coexist during execution.

5 Unified Metadata and Dataset Index

The first persistent index inspected in the notebook is `manifests/unified_subject_index.csv`. It contains **2,738 rows** and fields covering subject identity, dataset source/version, timepoints, modality availability, segmentation availability, label scheme, clinical metadata availability, longitudinal relationships, split information, lockbox state, overlap checks, site/scanner information, raw and processed paths, SHA-256 values, licensing/access metadata, QC state, exclusions, and notes.

Representative key fields include:

Field	Purpose in the project
<code>canonical_subject_id</code>	Canonical subject identifier used to align records across the project.
<code>dataset_source</code>	Identifies the originating dataset/cohort.

Field	Purpose in the project
timepoint_index, timepoint_label	Distinguish longitudinal scans belonging to the same patient.
T1_available, T1ce_available, T2_available, FLAIR_available	Record modality availability.
segmentation_available	Records whether a segmentation reference is available.
original_label_scheme	Describes the label convention used by the source dataset.
split, lockbox	Define dataset usage during development and final evaluation.
sha256_*	Provide file-integrity fingerprints for the modalities.
qc_status, qc_issues, excluded	Capture quality-control status and exclusion information.
raw_path, processed_path	Link metadata records to physical data locations.

At the beginning of the QC process, all 2,738 rows were marked pending; no rows were excluded at that stage.

6 File-Level Quality Control

6.1 BraTS 2021 Integrity Check

The notebook extracts the BraTS 2021 training archive, enumerating 7,507 archive members. For the 1,251 BraTS 2021 subjects, the QC script generates **6,255 file-level checks**, corresponding to five relevant imaging/segmentation files per subject: T1, T1ce, T2, FLAIR, and segmentation.

The QC result was:

	Expected files	Passed	Problem files
Existence	6,255	6,255	0
SHA-256 match	6,255	6,255	0
Readable NIfTI header	6,255	6,255	0

All 6,255 BraTS 2021 files therefore passed the three recorded checks. The raw files were reported as shape $240 \times 240 \times 155$, spacing $1 \times 1 \times 1$ mm, and LPS orientation before standardization.

6.2 Other Cohorts

A second QC pass uses dataset-specific manifests for BraTS-Africa, RHUH, MU, and UTSW. UTSW was not extracted at the time of this audit, so it was explicitly skipped by the file-checking code.

Dataset	Manifest records	Checked	Readable	Target shape observed
BraTS-Africa	731	730	730	$240 \times 240 \times 155$
MU-Glioma-Post	2,981	2,978	2,978	$240 \times 240 \times 155$
RHUH-GBM	721	720	720	Mostly $240 \times 240 \times 155$
UTSW-Glioma	625	0	–	Not extracted in this audit

The non-BraTS2021 audit flagged five problem files. Shape/orientation inspection additionally identified six RHUH scans belonging to RHUH-0028 with raw shape $230 \times 230 \times 138$, while the majority target was $240 \times 240 \times 155$. The same inspection also identified non-majority orientations in selected BraTS-Africa and RHUH scans. These findings are consistent with the need for a canonical standardization stage rather than assuming that all source datasets already share one geometry.

6.3 Intensity Inspection

The notebook samples 30 scans per dataset/modality and compares non-background intensity statistics. The reported medians demonstrate substantial scale differences across cohorts. For example, median T1 intensity values in the sampled data were approximately 400.5 for BraTS 2021, 453.0 for BraTS-Africa, 250.45 for MU, and -6.73 for RHUH. Similar cross-dataset differences were observed for T1ce, T2, and FLAIR.

These measurements provide a direct project-specific motivation for performing intensity normalization before model training.

7 Segmentation Labels and Label Harmonization

The segmentation masks use integer labels. The project documentation records the semantic convention used by the team as:

Label	Meaning
0	Background
1	Necrotic / necrotic component
2	Edema
3	Enhancing component
4	Resection cavity in datasets where this label is present

The notebook explicitly tests label remapping. The saved pipeline uses:

Listing 2: Configured segmentation label remapping

```

1 LABEL_MAP = {
2     'brats2021': {4: 3},
3     'utsw': {4: 3}
4 }
5 # Africa, RHUH, and MU keep their source labels in the saved pipeline.

```

For BraTS 2021, a representative mask changed from label set $\{0, 1, 2, 4\}$ to $\{0, 1, 2, 3\}$. The corresponding voxel counts in the notebook were:

Label	Before	After
0	8,870,695	8,870,695
1	11,738	11,738
2	12,836	12,836
4	32,731	0
3	0	32,731

For the tested MU example, the label 4 values were intentionally retained because the saved mapping does not remap MU labels.

The project therefore distinguishes between **label semantics** and **geometric/intensity harmonization**. The latter is applied broadly, while label harmonization follows the dataset-specific mapping encoded in the pipeline.

8 Longitudinal Structure and Subject Identity

Several cohorts contain multiple scans for the same patient over time. Consequently, the project does not identify a record by subject ID alone. The effective row identity is represented by the combination of `canonical_subject_id` and `timepoint_label`.

Examples recorded in the notebook include multiple RHUH timepoints per patient and multiple MU timepoints. The notebook verifies that there are **0 repeated rows** for the combined `canonical_subject_id + timepoint_label` key.

This distinction is important because a model-development dataset must preserve the longitudinal structure and avoid accidentally treating multiple scans from the same patient as independent subjects when creating data splits.

9 Missing Segmentations and Dataset-Specific Exceptions

The notebook identifies two MU timepoints without segmentation files:

- PatientID_0187, Timepoint_3 (reported in the final index as tp02).
- PatientID_0191, Timepoint_1 (reported in the final index as tp00).

The team handoff explicitly instructs that these cases should be skipped by the segmentation loader when a ground-truth mask is required.

The final v4 index also records 625 pending rows. These correspond to UTSW, which is kept as an external cohort in the split definition and is explicitly noted as not yet extracted/standardized in the final pipeline state.

The project therefore distinguishes between:

1. data that has been standardized and is available under `/content/processed`,
2. data that exists in the project repository but lacks a segmentation file for some timepoints, and
3. external data intentionally retained outside the routine training/validation regeneration path.

10 Duplicate and Overlap Audit

The notebook performs a SHA-256 based overlap analysis across the T1, T1ce, T2, and FLAIR files. Subjects sharing identical image hashes are grouped before splitting so that identical image content cannot be separated into different model-development splits.

The audit found **7 multi-subject groups**:

- 6 groups within MU.
- 1 group within BraTS-Africa.

Representative examples recorded by the notebook include MU pairs sharing 12, 8, or 4 files, and one BraTS-Africa pair sharing 4 files. These groups were treated as indivisible units during split generation.

11 Locked Dataset Split

The project constructs the split using grouped patient/image relationships and a fixed random seed of **42**. The primary datasets are BraTS2021, MU, and RHUH; BraTS-Africa and UTSW are assigned the external role.

The resulting row-level split is:

Dataset	Train	Validation	Test	External
BraTS2021	875	188	188	0
MU	405	81	110	0
RHUH	84	18	18	0
BraTS-Africa	0	0	0	146
UTSW	0	0	0	625

The corresponding patient-level counts are:

Dataset	Train	Validation	Test	External
BraTS2021	875	188	188	0
MU	142	31	30	0
RHUUH	28	6	6	0
BraTS-Africa	0	0	0	146
UTSW	0	0	0	625

The split is stored as `qc/split_v1.csv` and is marked as locked. A subsequent row-by-row comparison confirmed that `split_v1.csv` and `unified_subject_index_v3.csv` have identical subject keys, split assignments, lockbox flags, and canonical subject IDs.

Leakage-control rule: the saved preprocessing pipeline does not regenerate test or external data by default. These cohorts remain reserved for final evaluation, consistent with the team handoff instructions.

12 Standardization Design

The core preprocessing goal is to place heterogeneous source images into a common representation. The saved pipeline in `pipeline/standardize.py` implements four central transformations.

12.1 1. Canonical RAS Orientation

Each NIfTI image is loaded through `nibabel` and converted to the closest canonical orientation using `nib.as_closest_canonical(...)`, yielding a common RAS orientation in the processed output.

12.2 2. Common Target Shape

The target spatial shape is:

$$\mathbf{T} = (240, 240, 155).$$

If a source volume is smaller than the target along any axis, the pipeline pads symmetrically as evenly as possible before and after the existing volume. If a volume is larger than the target, the pipeline returns a failure for that record rather than silently cropping it.

The affine translation is adjusted after padding so that the spatial coordinate system remains consistent with the new array origin.

12.3 3. Brain-Region Z-Score Normalization

For non-segmentation modalities (T1, T1ce, T2, FLAIR), the pipeline estimates the background as the median voxel value, defines the brain region as the voxels differing from that background, and computes:

$$z = \frac{x - \mu}{\sigma + 10^{-8}},$$

where μ and σ are calculated from the selected non-background voxels.

The background is then written as zero. This converts each imaging modality to an approximately zero-mean, unit-standard-deviation representation inside the retained brain region.

12.4 4. Segmentation Handling

For segmentation maps, the pipeline avoids intensity normalization. Instead, values are rounded to integer labels, the configured dataset-specific label remapping is applied, and the resulting array is stored using an unsigned integer representation.

13 Standardization Pipeline Implementation

The reusable implementation is stored at:

Listing 3: Saved pipeline location

```
1 /content/drive/MyDrive/BrainMRI_Data/pipeline/standardize.py
```

The pipeline defines:

- a persistent project root on Drive,
- a temporary extraction area at /content/data,
- a temporary processed-output area at /content/processed,
- the target shape $240 \times 240 \times 155$,
- the kept scan types T1, T1ce, T2, FLAIR, and SEG,
- a list of TAR archives associated with the supported datasets,
- a resumable extraction mechanism, and
- multi-threaded file processing through `ThreadPoolExecutor`.

The extraction function creates completion markers such as `.done_brats2021`. If a dataset has already been extracted, the pipeline skips extraction rather than repeating it. Similarly, if an output file already exists, the standardization function returns `skipped` rather than recomputing the result.

The main entry point is:

Listing 4: Saved regeneration entry point

```
1 def regenerate(splits=('train', 'val'),
2               datasets=('brats2021', 'mu', 'rhuh')):
3     # test and external stay locked
4     idx = pd.read_csv(BASE / 'qc/unified_subject_index_v4.csv')
5     idx = idx[idx['split'].isin(splits)]
```

```

6         & idx['dataset_source'].isin(datasets)]
7     for ds in datasets:
8         ensure_extracted(ds)
9     return run(list(datasets),
10               subjects=set(idx['original_subject_id']))

```

This design makes the standardization step resumable and makes the train/validation restriction explicit in code rather than relying only on informal instructions.

14 Development-Time Validation of the Standardization Logic

Before/alongside the full preprocessing run, the notebook validates the transformation logic on representative samples.

For representative BraTS 2021, BraTS-Africa, and RHUH cases, the notebook reports:

- output orientation = RAS,
- output shape = $240 \times 240 \times 155$,
- imaging-volume mean approximately 0 and standard deviation approximately 1 within the non-zero brain region,
- segmentation label sets consistent with the applicable mapping, and
- tumor voxels retained inside the brain volume.

For the representative RHUH-0028 case, the tumor-inside-brain ratio was reported as **0.998** both before and after standardization, providing a concrete check that the geometric transformation did not move the segmentation outside the intended brain region.

15 Full Standardization Run

The notebook then executes the full processing stage into `/content/processed`. The recorded result is:

Metric	Recorded value
Files scheduled for processing	10,563
Status = done	10,363
Status = skipped	200
Status = failed	0 reported in the summary
Processed size	36.26 GB

The processed file count is consistent with the standardized cohorts and retained scan types:

Dataset	Timepoints / rows	Processed files
BraTS2021	1,251	6,255
BraTS-Africa	146	730
MU	596	2,978
RHUUH	120	600
Total	2,113	10,563

The MU total is reduced by two missing segmentation files, so not every timepoint contains all five retained files.

16 Processed-Data Audit

After standardization, the notebook audits every processed NIfTI file for shape and orientation. The result is:

Shape	Orientation	Files
(240, 240, 155)	RAS	10,563

A 300-image sample is then used to verify the normalized intensity behavior. Across the four standardized cohorts sampled, the reported mean values are approximately 0 and the reported standard deviations are approximately 1, as expected from the z-score transformation.

The segmentation audit also confirms that label sets are dataset-dependent. In the standardized output, BraTS-Africa, BraTS2021, MU, and RHUUH exhibit combinations of the expected integer labels, with MU retaining cases that include label 4 because the saved pipeline does not remap MU.

The audit is persisted to:

Listing 5: Persisted QC artifacts

```
1 BrainMRI_Data/qc/processed_audit.csv
2 BrainMRI_Data/qc/processing_log.csv
```

17 Unified Subject Index v4

Following successful processing, the notebook constructs `qc/unified_subject_index_v4.csv`. This version adds/updates the processed path information and records QC state based on the presence of standardized output folders.

The recorded result is:

v4 status	Rows
QC passed / processed path resolved	2,113
Pending	625

The two MU timepoints without segmentation are explicitly noted, and the notebook also records duplicate-image notes for 48 rows. The split and lockbox columns are checked against `split_v1.csv` before v4 is written, preventing accidental changes to the locked dataset allocation.

The 625 pending rows correspond to the UTSW external cohort that is explicitly documented as not yet extracted/standardized in the saved pipeline state.

18 Reproducibility and Consistency Check

The final notebook cell loads the saved pipeline, writes its outputs into a temporary comparison directory, and compares the resulting NIfTI arrays and affine matrices against the previously generated processed data.

The recorded validation was:

Comparison metric	Result
Files compared	40
Mismatches	0

This check is important because it tests not only that the code executes, but that the saved pipeline reproduces the already generated processed outputs at the array and affine level for the tested subset.

19 M1-T05: Model Input Pipeline, Smoke Training, and Experiment Tracking

M1-T05 extends the completed preprocessing layer into the first executable model-input and training path. The implementation is intentionally a **smoke test**, not a final training experiment. Its purpose is to prove that the cleaned MRI files produced by M1-T04 can be loaded by a segmentation model with the intended tensor shapes, label representation, augmentation logic, validation procedure, and experiment logging.

19.1 1. Project Package and Loader Structure

The notebook recreates the project package under `/content/repo` and defines the main modules used by the smoke test:

- `bmts.common.constants`: canonical MRI modality order and tumor-region definitions.

- `bmts.data.augmentation.transforms`: MONAI loading, region conversion, patch sampling, and augmentation transforms.
- `bmts.data.datasets.brats`: split-aware case discovery and DataLoader construction.
- `scripts/smoke_train.py`: executable end-to-end smoke-training script.
- `configs/data/loader_v1.yaml`: data-loader and augmentation configuration.

The configured imaging input is $T1, T1ce, T2, FLAIR$, stacked into four model channels. The segmentation target is converted to three binary region channels in the order TC, WT, ET .

19.2 2. Data Loading and Label Representation

The loader reads `t1.nii.gz`, `t1ce.nii.gz`, `t2.nii.gz`, `flair.nii.gz`, and `seg.nii.gz`. The expected tensors are:

$$image \in \mathbb{R}^{B \times 4 \times H \times W \times D}, \quad label \in \{0, 1\}^{B \times 3 \times H \times W \times D}.$$

The segmentation conversion is implemented through `SegToRegionsd`. Label 4 (resection cavity) is excluded from the three tumor-region outputs. The M1-T05 configuration trains on **BraTS2021 only**; MU and RHUH are not part of this smoke-training run.

19.3 3. Training Patch and Augmentation Strategy

Training uses $128 \times 128 \times 128$ patches. Tumor-centred sampling uses a positive-to-negative ratio of 2:1. The configured augmentations are:

Augmentation	Configuration
Random flip	probability 0.5 per axis
Random affine	probability 0.2
Rotation	up to 15 degrees
Scaling	0.1 range
Intensity scale/shift	probability 0.5
Gamma adjustment	probability 0.15
Elastic deformation	disabled in this smoke test

The training configuration keeps normalization disabled because M1-T04 already z-scored the standardized MRI modalities.

19.4 4. Validation Strategy

Validation uses whole standardized volumes and MONAI sliding-window inference with the same 128^3 ROI and 0.25 overlap. The locked test and external cohorts are not read by this training path. The loader also contains an explicit `LockboxError` guard against accidental use of the locked test split outside final evaluation.

19.5 5. Model Architecture and Optimization

The smoke test instantiates a MONAI 3D U-Net with four input channels and three output channels, channel widths (16, 32, 64, 128), three downsampling strides of 2, and one residual unit per stage. The loss is Dice loss with sigmoid outputs, and optimization uses Adam with learning rate 10^{-3} . Batch size is 2 and the smoke run is limited to two epochs. Seed 42 is configured.

19.6 6. Smoke-Test Execution Result

The executed run used:

Setting	Recorded value
Training patients	16
Validation patients	4
Epochs	2
Patch size	$128 \times 128 \times 128$
Batch size	2
Workers	2
Device	CPU
First-batch image tensor	(2, 4, 128, 128, 128)
First-batch label tensor	(2, 3, 128, 128, 128)
Smoke-test status	PASSED

The recorded epoch outputs were:

Epoch	Mean loss	TC Dice	WT Dice	ET Dice
1	0.9353	0.0120	0.0336	0.0066
2	0.9257	0.0295	0.0586	0.0133

These Dice values are **not final model-performance results**. They come from only two epochs on a tiny subset and are recorded here only as evidence that the complete model path executed successfully.

19.7 7. MLflow Experiment Tracking

The smoke run is tracked under the experiment BrainMRI_3D_UNet. The persistent SQLite backend is:

```
1 /content/drive/MyDrive/BrainMRI_Data/mlflow.db
```

The executed run ID is:

```
1 215715ab9fa8422092a767175f00e04d
```

The run finished successfully. Parameters include training/validation counts, epochs, batch size, ROI size, workers, learning rate, model name, modality and region order, split file, and processed-data root. Per-epoch training loss and validation Dice values are logged. The run also stores the JSON smoke report and `smoke_model.pt` checkpoint as artifacts.

19.8 8. Segmentation Metric Implementation and Verification

The notebook implements Dice and 95th-percentile Hausdorff distance (HD95). Dice is defined as:

$$Dice(P, G) = \frac{2|P \cap G|}{|P| + |G|}.$$

The implementation returns 1.0 when both binary masks are empty. For HD95, both surfaces are compared bidirectionally and the 95th percentile of the resulting distances is reported. If both masks are empty, the result is 0; if exactly one is empty, NaN is returned, matching the MONAI behavior tested.

The custom implementation was checked against MONAI on synthetic cases and on a real standardized BraTS2021 segmentation file. The real-data comparison produced:

Region	Custom Dice	MONAI Dice	Custom HD95	MONAI HD95
WT	0.9982	0.9982	0.0000	0.0000
TC	0.9989	0.9989	0.0000	0.0000
ET	0.9969	0.9969	0.0000	0.0000

A separate edge-removal test was also executed and the custom implementation matched MONAI. The current verified datasets record 1.0 mm voxel spacing in QC, so the smoke metric test used (1, 1, 1) mm. The standardization code pads arrays but does not perform voxel-spacing resampling; future evaluations should read actual NIfTI spacing from the header when heterogeneous spacing is possible.

19.9 9. M1-T05 Review Status

Component	Status	Evidence
DataLoader construction	Passed	16 train + 4 validation patients loaded successfully.
Tensor shape checks	Passed	4-channel image and 3-channel label tensors verified.
Finite-value / binary-label checks	Passed	No NaN/Inf; labels asserted binary.
3D U-Net forward/backward	Passed	Two training epochs completed without failure.
Validation inference	Passed	Sliding-window validation completed.
MLflow tracking	Passed	Run FINISHED with parameters, metrics, report, and model artifact.
Dice implementation	Verified	Matched MONAI on synthetic and real-data tests.
HD95 implementation	Verified	Matched MONAI on synthetic and real-data tests.
Final model training	Not completed	M1-T05 is explicitly a smoke test.
Final test/external evaluation	Not performed	Locked cohorts remain reserved.

19.10 10. Technical Limitations and Follow-Up Notes

1. The observed Dice values are smoke-test values only and are not a meaningful estimate of final model quality.
2. The run used CPU because no CUDA GPU was available in the recorded Colab session; the timings are environment-specific.
3. The current loader configuration demonstrates only BraTS2021 training. Multi-dataset training is not demonstrated by this notebook.
4. The longitudinal case-discovery helper is safe for the current single-timepoint BraTS2021 smoke test; a future multi-timepoint configuration should bind each split row explicitly to its intended timepoint.
5. HD95 should use each case's actual NIFTI voxel spacing in production if any future cohort differs from the verified 1 mm spacing.
6. The SQLite MLflow backend is suitable for this single-session smoke experiment but is not a shared remote tracking server for concurrent multi-user experimentation.

20 Project Artifacts Produced

The milestone produces or updates the following persistent project artifacts:

Artifact	Purpose
manifests/unified_subject_index.csv	Initial unified subject/timepoint metadata index.
qc/brats2021_scan.csv	BraTS 2021 file-level integrity and geometry audit.
qc/other_datasets_scan.csv	File-level audit for Africa, MU, RHUH, and UTSW status.
qc/unified_subject_index_v2.csv	QC-updated index after BraTS 2021 validation.
qc/unified_subject_index_v3.csv	Index containing locked split assignments.
qc/split_v1.csv	Persisted split/lockbox definition (seed 42).
qc/unified_subject_index_v4.csv	Final milestone index with processed paths and QC state.
qc/processing_log.csv	Per-file standardization status and errors.
qc/processed_audit.csv	Post-standardization shape/orientation/intensity/label audit.
pipeline/standardize.py	Reusable standardization and regeneration pipeline.
pipeline/metrics.py	Persisted Dice and HD95 evaluation functions verified against MONAI.
runs/M1-T05_smoke_mlflow/smoke_report.json	JSON report produced by the model smoke run.
runs/M1-T05_smoke_mlflow/smoke_model.pt	Model checkpoint produced by the smoke run.
mlflow.db	Persistent SQLite MLflow tracking database for the M1-T05 experiment.
/content/processed/	Colab runtime directory containing standardized MRI volumes.

21 Operational Notes for Team Use

The accompanying team handoff specifies the following operational procedure for another team member who needs the processed data in a new Colab session:

1. Add a shortcut to the shared BrainMRI_Data folder in Google Drive.

2. Mount Google Drive in Colab.
3. Add `/content/drive/MyDrive/BrainMRI_Data/pipeline` to Python's module search path.
4. Load `standardize.py`.
5. Run `sm.regenerate()`.

The team handoff estimates roughly one to two hours for the standardization run depending on the Colab environment and notes that approximately 60 GB of free runtime storage may be required. Processed images are written to `/content/processed` rather than persisted to Drive, so the generated outputs are tied to the active Colab runtime unless separately copied elsewhere.

The regeneration logic is designed to resume: previously produced files are detected and skipped rather than reprocessed.

21.1 Recommended command sequence

Listing 6: Minimal regeneration sequence from the team handoff

```
1 from google.colab import drive
2
3 drive.mount('/content/drive')
4
5 import sys
6 sys.path.insert(0, '/content/drive/MyDrive/BrainMRI_Data/pipeline')
7
8 import standardize as sm
9 log = sm.regenerate()
```

22 Milestone 1 Status Summary

Area	Status	Evidence recorded in the notebooks
Dataset organization	Completed	Persistent BrainMRI_Data structure with raw/manifests/qc/pipeline.
Metadata consolidation	Completed	2,738-row unified index and subsequent v2/v3/v4 versions.
BraTS 2021 QC	Completed	6,255/6,255 files existed, matched checksum, and were readable.
Other-cohort QC	Completed for extracted cohorts	Africa/MU/RHUU checked; UTSW was not extracted in the recorded audit.
Overlap audit	Completed	7 duplicate-image groups identified and preserved during splitting.
Locked split	Completed	split_v1.csv, seed 42, test/external lockbox.
Standardization logic	Completed	Saved as pipeline/standardize.py.
Full standardization	Completed in recorded notebook run	10,563 processed files; 10,363 done; 200 skipped.
Processed audit	Completed	10,563 files at target shape and RAS orientation.
Reproducibility check	Passed on tested subset	40 files compared; 0 mismatches.
M1-T05 data loader	Passed (smoke test)	16 train + 4 validation BraTS2021 patients loaded and batched successfully.
M1-T05 model path	Passed (smoke test)	3D U-Net completed 2 epochs and validation without runtime failure.
MLflow tracking	Passed	Run FINISHED with parameters, epoch metrics, report, and checkpoint artifacts.
Metric implementation	Verified	Dice and HD95 matched MONAI on tested synthetic and real-data cases.
Final model training	Pending	Only a 2-epoch smoke test has been executed.
UTSW integration	Pending	625 external rows remain pending; saved regenerate() excludes UTSW by default.

23 Conclusion

Milestone 1 established the project data pipeline required before model development. The work moved from raw heterogeneous MRI collections to a controlled and documented processing environment with explicit metadata, file integrity checks, longitudinal subject handling, duplicate-image grouping, locked data splits, reproducible standardization logic, and post-processing audits.

The most important technical result is that preprocessing is no longer an ad-hoc sequence of notebook cells. It has been consolidated into a reusable `standardize.py` pipeline whose regeneration function reads the locked v4 index and processes only the intended development cohorts by default. The resulting processed data has a consistent spatial representation and normalized imaging intensities, while the segmentation masks are preserved as discrete label volumes.

The milestone therefore provides a stable foundation for the next project phase: constructing the model input loader, selecting and implementing the segmentation architecture, training on the training split, monitoring validation performance, and reserving the locked test/external cohorts for final evaluation.

Final Milestone 1 statement: the project has established and validated the data-preparation layer and has demonstrated, through M1-T05, that the standardized MRI data can pass through the intended loader, augmentation, 3D U-Net, validation, metric, and MLflow tracking path. This milestone is not a claim of final model quality; full training and locked-cohort evaluation remain subsequent tasks.

A Key Terms Used in the Project

Term	Meaning in this project
MRI	Magnetic Resonance Imaging volumes used as model input.
T1 / T1ce / T2 / FLAIR	MRI modalities retained by the standardization pipeline.
SEG	Segmentation mask containing integer labels for tumor-related regions.
NIfTI / .nii.gz	File format used for volumetric neuroimaging data.
QC	Quality control: file existence, checksum, readability, geometry, and related checks.
SHA-256	Cryptographic file fingerprint used to verify that files match expected contents.
RAS / LPS	Canonical anatomical orientation codes used for image geometry.
Manifest	Structured metadata describing available scans and their properties/paths.

Term	Meaning in this project
Split	Assignment of data to train, validation, test, or external groups.
Lockbox	Data deliberately reserved from routine development and preprocessing regeneration.
Timepoint	A longitudinal scan instance associated with a patient.
Pipeline	Ordered, reusable sequence of preprocessing operations.

B Evidence Basis

This report is based on two project-provided sources:

1. The supplied notebook `Copy of brain_mri_datasets_kaggle.ipynb`, including its executed outputs where present.
2. The supplied notebook `MLflow_Training.ipynb`, including its executed smoke-training, MLflow, and metric-validation outputs.
3. The team handoff message describing the saved pipeline, Drive layout, processing behavior, storage requirements, label conventions, split restrictions, and known dataset-specific exceptions.

Where the notebook and the handoff distinguish between exploratory acquisition and the final saved regeneration path, the report preserves that distinction rather than merging the two stages into a single claim.